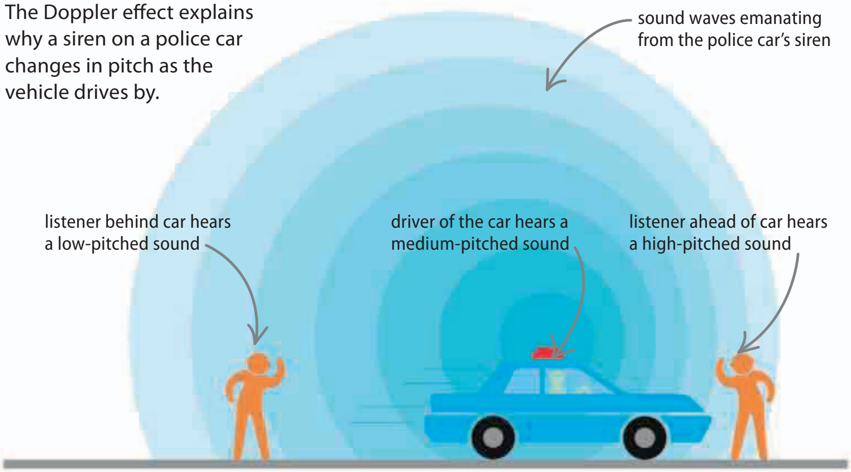
Doppler effect

If a sound source is moving toward a listener, the pulses of pressure that make up the sound waves get closer together because the source is moving a little closer to each one before sending out the next. This means that the sound's frequency is higher than if the source were stationary. If the source is receding, the pulses become farther apart and the frequency lowers. This is called the Doppler effect.

▽ **Police siren**
The Doppler effect explains why a siren on a police car changes in pitch as the vehicle drives by.



REAL WORLD

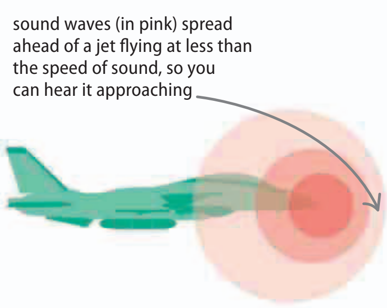
Sound underwater

Sound travels better in water than in air. Marine animals use sound for a wide range of tasks. Some use it to communicate over huge distances, others to probe their surroundings, while some even use it to stun their prey. Dolphins and some whale species are especially dependent on sound for communication, which makes them particularly vulnerable to underwater noise pollution.

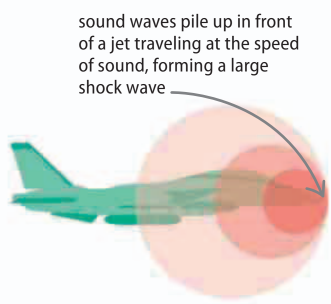
Supersonic motion

Sound travels at a speed of around 343 m (1,340 ft) per second through air. However, when an object travels faster than sound, it overtakes the sound waves ahead of it. An example of this is the supersonic jet, which flies faster than the speed of sound, so a person cannot hear it coming toward him or her—the jet passes before the sound arrives. However, when the sound catches up, it arrives suddenly as a shock wave, which is heard on the ground as a sonic boom.

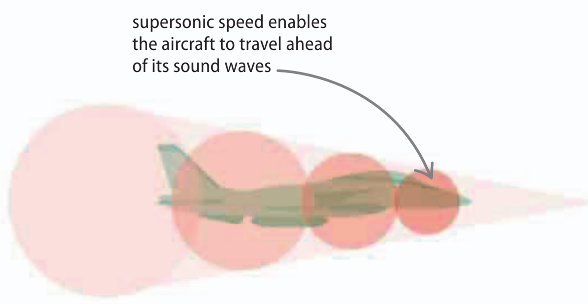
High-power, high-frequency sound can be used to **smash kidney stones apart**, avoiding the need for surgery.



△ **Subsonic flight**
The sound waves ahead of an aircraft flying slower than the speed of sound have higher frequencies than those behind them.



△ **The shock front**
When the speed of sound is reached, the sound waves can no longer spread ahead of the plane, creating a shock front.



△ **Supersonic flight**
The shock front of a supersonic plane is heard as a sonic boom by anyone it passes over.